

README

Replication Package for “The Geographic Footprint of U.S. Dairy Policy”

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1 OVERVIEW

The code and data contained in this package allow one to replicate the results in “The Geographic Footprint of U.S. Dairy Policy” (*Quantitative Economics*) by Hanon, Mérel, and Sumner.

The package contains R code to construct the data files needed to run the main GAMS code from the raw data files, the GAMS code itself, and additional R code that turns the csv files generated by the GAMS code into text files. Each text file corresponds to a table from the main text or the Supplemental Appendix. There are 7 results tables in the main text labeled from Table 1 to Table 7, and additional results tables in the Supplemental Appendix (Tables D.1–D.4 and Table E.1). The package also contains Mathematica code that produces Figure 1 in the main text and the figures in Supplemental Appendix G, as well as a handful of numbers used to substantiate a claim made in Supplemental Appendix G. Figure A.1 in Supplemental Appendix A is generated from a separate R code file.

Finally, the GAMS code produces numbers that are referred to in various sections of the main text and the Supplemental Appendix and that are not part of a table. It also produces numbers that are used to substantiate various claims regarding results not directly reported in a table.

The code is executed using a single shell script, `run_all.sh`. This script includes code that will automatically install all necessary R packages using the `{renv}` package. As a result, the first run of the code will take longer than subsequent runs. When last run on a computer running Windows, the code took approximately 3 minutes to execute. When last run on a computer running Mac OS, the code took approximately 10 minutes to execute.

1.1 DESCRIPTION OF PACKAGE CONTENTS

The top-level directory of this package, titled `Geographic-Footprint-US-Dairy`, contains several subfolders and the following files:

- This `README` file (in both `.pdf` and `.md` format).
- `README.qmd`, a Quarto file used to generate this `README` (which also uses some files in the `.quarto` subfolder)
- `LICENSE.txt`, which contains the text of the Apache 2.0 license used for the code in this replication package and the Open Data Commons Public Domain Dedication and License v1.0 used for the data in this replication package.
- `Geographic-Footprint-US-Dairy.Rproj`, a file designating this directory as an RStudio Project.
- `.Rprofile`, a file containing code that will be executed when R loads. Specifically, the R script `activate.R` in the `renv` is executed to enable the `{renv}` R package and allow the necessary environment to be replicated. See Section 6 or the [{renv} documentation](#) for more details.
- `renv.lock`, a file with metadata for each R package used in this project. See the [{renv} documentation](#) for more details.

The subfolders contained within the top-level directory include:

- **Data:** Includes raw data in `Raw` subfolder, `Lookup_Tables` subfolder, and `Other_Sources` subfolder with PDF copies of specific data sources. Additionally, some data are saved in a `Processed` subfolder created during data processing.
- **GAMS CODE:** Contains the main model code to be executed in GAMS Studio. Final data for use in the model are saved in a subfolder called `CSV DATA FILES` during processing.
- **MATHEMATICA CODE:** Contains the code for Wolfram Mathematica used to produce figures.
- **R CODE:** Contains R code used for processing raw data for use in the main model (`Data-Processing` subfolder) and for processing model results into tables (`Tables` subfolder), figures (`Figures` subfolder), and in-text numerical claims (`Claims` subfolder).
- **References:** Contains `.bib` files for data and R package citations as well as a `.csl` file.
- **renv:** Contains R code and settings files related to `{renv}` package management. When `renv::restore` is run, a `library` subfolder will be created to contain all of the R packages used in this project.

The directory tree below outlines the top two levels of the project directory.

```

../Geographic-Footprint-US-Dairy
├── .Rprofile
├── Data
│   ├── Lookup_Tables
│   ├── Other_Sources
│   └── Raw
├── GAMS CODE
│   ├── FMMO_Model_QE.gms
│   └── FMMO_Model_QE.gsp
├── Geographic-Footprint-US-Dairy.Rproj
├── LICENSE.txt
├── MATHEMATICA CODE
│   ├── FMMOQE.nb
│   └── FMMOQE.wls
├── R CODE
│   ├── Claims
│   ├── Data-Processing
│   ├── Figures
│   └── Tables
├── README.md
├── README.pdf
├── README.qmd
├── References
│   ├── data-citations.bib
│   ├── econometrica.csl
│   └── grateful-refs.bib
├── renv
│   ├── activate.R
│   ├── library
│   └── settings.json
├── renv.lock
└── run_all.sh

```

Additionally, several folders are created in the top-level directory to contain the finished tables, figures, and in-text numerical claims: `MANUSCRIPT CLAIMS`, `MANUSCRIPT FIGURES`, and `MANUSCRIPT TABLES`.

2 DATA AVAILABILITY STATEMENTS

All of the data we used in this analysis are public data originally produced by either state or federal government sources. In most cases, we accessed and downloaded the data through online tools. We walk through the steps to access each data source collected this way in the following sections. For a few files, we obtained the raw data directly from the government agencies that produced the data. Table 1 lists the data files in the [Data/Raw](#) folder, all of which are provided in this replication package.

2.1 STATEMENT ABOUT RIGHTS

- ☒ We certify that the authors of the manuscript have legitimate access to and permission to use the data used in this manuscript.
- ☒ We certify that the authors of the manuscript have documented permission to redistribute/publish the data contained within this replication package. Appropriate permission are documented in the [LICENSE.txt](#) file.

2.2 STATEMENT OF AVAILABILITY

- ☒ All data are publicly available.

Table 1 lists each data file in the [Data/Raw](#) folder and states their availability.

2.3 LICENSE FOR DATA

The data included in this replication package are made available under the Public Domain Dedication and License v1.0 whose full text can be found at: <http://opendatacommons.org/licenses/pddl/1.0/>

See [LICENSE.txt](#) for details.

2.4 USDA AGRICULTURAL MARKETING SERVICE DATA

We used Federal Milk Marketing Order (FMMO) data from the USDA Agricultural Marketing Service (AMS). We accessed and downloaded most of the AMS data through the [Datamart](#) service.

Producer Milk by State of Origin

- Files:
 - [MilkAndMilkPooledByState2017.csv](#)
 - [MilkAndMilkPooledByState2019.csv](#)
- Source: USDA Agricultural Marketing Service
- Provided: Yes

These data show the total milk produced in each state and the quantity of milk originating in each state that was pooled in each FMMO. These files were provided to the authors by AMS in [.xlsx](#) format. The files included in this package were simply saved in [.csv](#) format but are otherwise unedited.

Producer Receipts and Component Tests

- Files:
 - [Datamart-Export_FM_Producer_Receipts_2017.csv](#)
 - [Datamart-Export_FM_Producer_Components-Butterfat_2017.csv](#)
 - [Datamart-Export_FM_Producer_Components-Protein_2017.csv](#)
 - [Datamart-Export_FM_Producer_Components-Other Solids_2017.csv](#)

From Datamart homepage select “FMMOS” from the Commodity menu. Click “Utilization Reports” and “Producer Milk Components”. Under “Sub Report Type to Include”, check the boxes for Receipts, Butterfat, Protein, and Other Solids (alternatively, uncheck the boxes for NFS and SomCell). Under Report Year, switch the dropdown menu from “All” to “Equal To” and enter 2017 in the box that appears. Click “Continue”. Select CSV as the file format and click “Generate Report” (U.S. Department of Agriculture, Agricultural Marketing Service (2017a).

Class I Utilization of Producer Milk

- Files:
 - [Datamart-Export_FM_ClassI_Util-Butterfat_2017.csv](#)
 - [Datamart-Export_FM_ClassI_Util-Milk_2017.csv](#)
 - [Datamart-Export_FM_ClassI_Util-NFS_2017.csv](#)
 - [Datamart-Export_FM_ClassI_Util-Utilization_2017.csv](#)

From Datamart homepage select “FMMOS” from the Commodity menu. Click “Utilization Reports” and “Class I Utilization of Producer Milk”. Under Report Year, switch the dropdown menu from “All” to “Equal To” and enter 2017 in the box that appears. Click “Continue”. Select CSV as the file format and click “Generate Report” (U.S. Department of Agriculture, Agricultural Marketing Service (2017b).

Class II Utilization of Producer Milk

- Files:
 - [Datamart-Export_FM_ClassII_Util-Butterfat_2017.csv](#)
 - [Datamart-Export_FM_ClassII_Util-NFS_2017.csv](#)
 - [Datamart-Export_FM_ClassII_Util-Utilization_2017.csv](#)

From Datamart homepage select “FMMOS” from the Commodity menu. Click “Utilization Reports” and “Class II Utilization of Producer Milk”. Under “Sub Report Type to Include”, uncheck the box for Milk. Under Report Year, switch the dropdown menu from “All” to “Equal To” and enter 2017 in the box that appears. Click “Continue”. Select CSV as the file format and click “Generate Report” (U.S. Department of Agriculture, Agricultural Marketing Service (2017c).

Class III Utilization of Producer Milk

- Files:
 - [Datamart-Export_FM_ClassIII_Util-Butterfat_2017.csv](#)
 - [Datamart-Export_FM_ClassIII_Util-Other Solids_2017.csv](#)
 - [Datamart-Export_FM_ClassIII_Util-Protein_2017.csv](#)
 - [Datamart-Export_FM_ClassIII_Util-Utilization_2017.csv](#)

From Datamart homepage select “FMMOS” from the Commodity menu. Click “Utilization Reports” and “Class III Utilization of Producer Milk”. Under “Sub Report Type to Include”, uncheck the box for Milk. Under Report Year, switch the dropdown menu from “All” to “Equal To” and enter 2017 in the box that appears. Click “Continue”. Select CSV as the file format and click “Generate Report” (U.S. Department of Agriculture, Agricultural Marketing Service (2017d).

Class IV Utilization of Producer Milk

- Files:
 - [Datamart-Export_FM_ClassIV_Util-Butterfat_2017.csv](#)
 - [Datamart-Export_FM_ClassIV_Util-NFS_2017.csv](#)
 - [Datamart-Export_FM_ClassIV_Util-Utilization_2017.csv](#)

From Datamart homepage select “FMMOS” from the Commodity menu. Click “Utilization Reports” and “Class IV Utilization of Producer Milk”. Under “Sub Report Type to Include”, uncheck the box for Milk. Under Report Year, switch the dropdown menu from “All” to “Equal To” and enter 2017 in the box that appears. Click “Continue”. Select CSV as the file format and click “Generate Report” (U.S. Department of Agriculture, Agricultural Marketing Service (2017e).

Advanced Prices

- Files:
 - [Datamart-Export_DY_Advanced_Prices_2017.csv](#)

From Datamart homepage select “Dairy” from the Commodity menu. Click “All Dairy” and “(DY_AD101) National Dairy Product Prices - Advanced”. Under Report Year, switch the dropdown menu from “All” to “Equal To” and enter 2017 in the box that appears. Click “Continue”. Uncheck all fields on the next screen and select “Base Class 1 Price”, “Base Skim Milk Class 1 Price”, and “Advanced Butterfat Factor”. Select CSV as the file format and click “Generate Report” (U.S. Department of Agriculture, Agricultural Marketing Service (2017f)).

Class Prices

- Files:
 - [Datamart-Export_DY_Class_Prices_2017.csv](#)

From Datamart homepage select “Dairy” from the Commodity menu. Click “All Dairy” and “(DY_CL102) National Dairy Product Prices - Monthly”. Under Report Year, switch the dropdown menu from “All” to “Equal To” and enter 2017 in the box that appears. Click “Continue”. Uncheck all fields on the next screen and select “Class 2 Price”, “Class 2 Butterfat Price”, “Class 3 Price”, “Class 3 Skim Milk Price”, “Class 4 Price”, “Class 4 Skim Milk Price”, “Butterfat Price”, “Nonfat Solids Price”, “Protein Price”, and “Other Solids Price”. Select CSV as the file format and click “Generate Report” (U.S. Department of Agriculture, Agricultural Marketing Service (2018)).

2.5 MILK POOLING DATA FROM STATE SOURCES

In addition to the FMMO data from AMS, we used data from two state milk marketing orders, the California Milk Marketing Order and the Western New York Milk Marketing Area. In both cases, we accessed PDF reports published by the respective state departments of agriculture and generated CSV files with the relevant data. The original PDF reports and the CSV files are both available in this replication package.

California Milk Pooling Data

- Files:
 - [California_Dairy_Statistics_Annual_2017.pdf](#)
 - [California_Statistics_2017.csv](#)
 - [California_Monthly_ClassI_2017.csv](#)
- Source: California Department of Food and Agriculture
- Original URL: https://www.cdfa.ca.gov/dairy/pdf/Annual/2017/2017_Statistics_Annual.pdf
- Archived URL: https://web.archive.org/web/20181130220414/https://www.cdfa.ca.gov/dairy/pdf/Annual/2017/2017_Statistics_Annual.pdf

We obtained data on milk volumes and utilization of components by class in the California State Milk Marketing Order from the California Dairy Statistics Annual report for 2017 published by the California Department of Food and Agriculture (California Department of Food and Agriculture (2017)). We created two CSV files with data from a PDF copy of the report, which itself is included in the [Data/Other_Sources](#) folder. The report is no longer accessible through the CDFA website, but can be accessed through the *Internet Archive*. An archival URL is included.

Western New York Milk Pooling Data

- Files:
 - [Western_New_York_Annual_Statistical_Report_2017.pdf](#)
 - [Western_New_York_Data_2017.csv](#)
- Source: New York Department of Agriculture and Markets
- Original URL: <https://agriculture.ny.gov/system/files/documents/2019/06/WNYAnnual2017.pdf>

We obtained data on milk volumes and utilization of components by class in the Western New York Milk Marketing Area from the Western New York Milk Marketing Area Annual Statistical Report 2017 published by the New York Department of Agriculture and Markets (New York Department of Agriculture and Markets (2017)). We create a CSV file with data from a PDF copy of the report, which itself is included in the [Data/Other_Sources](#) folder and can be accessed through the [New York Department of Agriculture and Markets website](#).

2.6 NATIONAL AGRICULTURAL STATISTICS SERVICE DATA

The National Agricultural Statistics Service (NASS) data we used fall into two general categories: Census data and Survey data. We used Census data from the 2017 Census of Agriculture (U.S. Department of Agriculture, National Agricultural Statistics Service (2019)). When specific data series were unavailable in the 2017 Census, or when data from years other than 2017 were needed, we used data from NASS survey reports. Those survey reports are cited individually below. In all cases, we used the [NASS Quick Stats](#) service to download data. The steps to access each raw data file are outlined below.

Milk Production

- Files:
 - [NASS_Milk_Production_2017.csv](#)

We used survey data on state-level milk production volumes from the *Milk Production* report (U.S. Department of Agriculture, National Agricultural Statistics Service (2018a)). To download these data from Quick Stats, select “MILK” from the Commodity box; select “PRODUCTION” from the Category box; select “MILK - PRODUCTION, MEASURED IN LB” from the Data Item box; select “STATE” from the Geographic Level box; select 2017 from the Year box; select “ANNUAL” from the Period Type box.

Dairy Products

- Files:
 - [NASS_Dairy_Products_2017.csv](#)

We used survey data on state and multi-state regional dairy product production volumes from the *Dairy Products Annual Summary* report (U.S. Department of Agriculture, National Agricultural Statistics Service (2018b)). To download these data from Quick Stats, select “SURVEY” from the Program box; select “ANIMALS & PRODUCTS” from the Sector box; select “DAIRY” from the Group box; select “BUTTER”, “CHEESE”, “CREAM”, “ICE CREAM”, “MILK”, “SHERBET”, “WHEY”, and “YOGURT” from the Commodity box; select “PRODUCTION” from the Category box; select all of the following from the Data Item box:

- BUTTER - PRODUCTION, MEASURED IN LB
- CHEESE - PRODUCTION, MEASURED IN LB
- CHEESE, COTTAGE, CREAMED, FAT CONTENT GE 4 PCT - PRODUCTION, MEASURED IN LB
- CHEESE, COTTAGE, LOWFAT, FAT CONTENT LT 4 PCT - PRODUCTION, MEASURED IN LB
- CREAM, SOUR - PRODUCTION, MEASURED IN LB
- ICE CREAM, LOWFAT - PRODUCTION, MEASURED IN GALLONS
- ICE CREAM, MIX, LOWFAT - PRODUCTION, MEASURED IN GALLONS
- ICE CREAM, MIX, REGULAR - PRODUCTION, MEASURED IN GALLONS
- ICE CREAM, REGULAR, HARD - PRODUCTION, MEASURED IN GALLONS
- MILK, DRY, NONFAT, HUMAN - PRODUCTION, MEASURED IN LB
- MILK, SKIM, UNSWEETENED, CONDENSED, BULK - PRODUCTION, MEASURED IN LB
- SHERBET, HARD - PRODUCTION, MEASURED IN GALLONS
- SHERBET, MIX - PRODUCTION, MEASURED IN GALLONS
- WHEY, DRY, HUMAN - PRODUCTION, MEASURED IN LB
- YOGURT, MIX, FROZEN - PRODUCTION, MEASURED IN GALLONS

- YOGURT, PLAIN & FLAVORED - PRODUCTION, MEASURED IN LB

Select “REGION: MULTI-STATE” and “STATE” from the Geographic Level box; select 2017 from the Year box; select “ANNUAL” from the Period Type box.

Total Cropland and Harvested Cropland Acreage

- Files:
 - [NASS_All_Cropland_2017.csv](#)
 - [NASS_All_Cropland_Harvested_2017.csv](#)

We used 2017 Census data on total cropland acreage and harvested cropland acreage (U.S. Department of Agriculture, National Agricultural Statistics Service (2019). To download these data from Quick Stats, for both files select “CENSUS” from the Program box; select “AG LAND” from the Commodity box; select “AREA” from the Category box.

For the file [NASS_All_Cropland_2017.csv](#), select “AG LAND, CROPLAND - ACRES” and “AG LAND, CROPLAND, PASTURED ONLY - ACRES” from the Data Item box. For the file [NASS_All_Cropland_Harvested_2017.csv](#), select “AG LAND, CROPLAND, HARVESTED - ACRES” from the Data Item box. For both files, select “TOTAL” from the Domain box; select “STATE” from the Geographic Level box; select 2017 from the Year box.

Feed Crop Acreage

- Files:
 - [NASS_Census_Feed_Crop_Acreage_2017.csv](#)

We used 2017 Census data on feed crop acres harvested (U.S. Department of Agriculture, National Agricultural Statistics Service (2019). To download these data from Quick Stats, select “CENSUS” from the Program box; select “FIELD CROPS” from the Group box; select “BARLEY”, “BUCKWHEAT”, “CANOLA”, “CORN”, “COTTON”, “FLAXSEED”, “HAY”, “HAYLAGE”, “MILLET”, “OATS”, “RICE”, “RYE”, “SORGHUM”, “SOYBEANS”, and “WHEAT” from the Commodity box; select “AREA HARVESTED” from the Category box; select all of the following from the Data Item box:

- BARLEY - ACRES HARVESTED
- BUCKWHEAT - ACRES HARVESTED
- CANOLA - ACRES HARVESTED
- CORN, GRAIN - ACRES HARVESTED
- CORN, SILAGE - ACRES HARVESTED
- COTTON - ACRES HARVESTED
- FLAXSEED - ACRES HARVESTED
- HAY - ACRES HARVESTED
- HAYLAGE - ACRES HARVESTED
- MILLET, PROSO - ACRES HARVESTED
- OATS - ACRES HARVESTED
- RICE - ACRES HARVESTED
- RYE - ACRES HARVESTED
- SORGHUM, GRAIN - ACRES HARVESTED
- SORGHUM, SILAGE - ACRES HARVESTED
- SOYBEANS - ACRES HARVESTED
- WHEAT - ACRES HARVESTED

Select “TOTAL” from the Domain box; select “STATE” from the Geographic Level box; select 2017 from the Year box.

Other Crop Acreage

- Files:
 - [NASS_Census_Other_Crop_Acreage_2017.csv](#)

We used 2017 Census data on other crop acres harvested (U.S. Department of Agriculture, National Agricultural Statistics Service (2019). To download these data from Quick Stats, select “CENSUS” from the Program box; select “FIELD CROPS”, “FRUIT & TREE NUTS”, and “VEGETABLES” from the Group box; select “BEANS”, “BERRY TOTALS”, “CHICKPEAS”, “GRASSES & LEGUMES TOTALS”, “HOPS”, “LENTILS”, “MINT”, “ORCHARDS”, “PEANUTS”, “PEAS”, “POPCORN”, “SAFFLOWER”, “SUGARBEETS”, “SUGARCANE”, “SUNFLOWER”, “TOBACCO”, and “VEGETABLE TOTALS” from the Commodity box; select “AREA BEARING & NON-BEARING”, “AREA GROWN”, and “AREA HARVESTED” from the Category box; select all of the following from the Data Item box:

- BEANS, DRY EDIBLE, (EXCL CHICKPEAS & LIMA) - ACRES HARVESTED
- BEANS, DRY EDIBLE, LIMA - ACRES HARVESTED
- BERRY TOTALS - ACRES GROWN
- CHICKPEAS - ACRES HARVESTED
- GRASSES & LEGUMES TOTALS, SEED - ACRES HARVESTED
- HOPS - ACRES HARVESTED
- LENTILS - ACRES HARVESTED
- MINT, OIL - ACRES HARVESTED
- ORCHARDS - ACRES BEARING & NON-BEARING
- PEANUTS - ACRES HARVESTED
- PEAS, DRY EDIBLE - ACRES HARVESTED
- POPCORN, SHELLS - ACRES HARVESTED
- SAFFLOWER - ACRES HARVESTED
- SUGARBEETS - ACRES HARVESTED
- SUGARCANE, SUGAR - ACRES HARVESTED
- SUNFLOWER - ACRES HARVESTED
- TOBACCO - ACRES HARVESTED
- VEGETABLE TOTALS, IN THE OPEN - ACRES HARVESTED

Select “TOTAL” from the Domain box; select “STATE” from the Geographic Level box; select 2017 from the Year box.

Crop Sales

- Files:
 - [NASS_Total_Crop_Sales_2017.csv](#)
 - [NASS_Crop_Sales_2017.csv](#)

We used 2017 Census data on total crop sales and individual crop sales (U.S. Department of Agriculture, National Agricultural Statistics Service (2019). Four commodities do not have sales reported in the Census of Agriculture, and therefore we used survey data on the value of production for these commodities: hay, peanuts, sugarbeets, and sugarcane. Quick Stats contains records for “PRODUCTION, MEASURED IN \$” that are a combination of production volumes reported in the *Crop Production Annual Summary* report and prices received reported in the *Agricultural Prices* report (National Agricultural Statistics Service (2018a); National Agricultural Statistics Service (2018b).

To download the data in the file [NASS_Total_Crop_Sales_2017.csv](#) from Quick Stats, select “CENSUS” from the Program box; select “CROP TOTALS” from the Group box; select “CROP TOTALS” from the Commodity box; select “SALES” from the Category box; select “CROP TOTALS - SALES, MEASURED IN \$” from the Data Item box; select “TOTAL” from the Domain box; select “STATE” from the Geographic Level box; select 2017 from the Year box.

To download the data in the file [NASS_Crop_Sales_2017.csv](#) from Quick Stats, select “FIELD CROPS” from the Group box; select “BARLEY”, “CORN”, “COTTON”, “FRUIT & TREE NUT TOTALS”, “HAY”, “PEANUTS”, “RICE”, “SORGHUM”, “SOY-BEANS”, “SUGARBEETS”, “SUGARCANE”, “TOBACCO”, “VEGETABLE TOTALS”, and “WHEAT” from the Commodity box; select “PRODUCTION” and “SALES” from the Category box; select all of the following from the Data Item box:

- BARLEY - SALES, MEASURED IN \$
- CORN - SALES, MEASURED IN \$
- COTTON, LINT & SEED - SALES, MEASURED IN \$
- FRUIT & TREE NUT TOTALS - SALES, MEASURED IN \$
- GRAIN, OTHER - SALES, MEASURED IN \$
- HAY - PRODUCTION, MEASURED IN \$
- PEANUTS - PRODUCTION, MEASURED IN \$
- RICE - SALES, MEASURED IN \$
- SORGHUM - SALES, MEASURED IN \$
- SOYBEANS - SALES, MEASURED IN \$
- SUGARBEETS - PRODUCTION, MEASURED IN \$
- SUGARCANE, SUGAR & SEED - PRODUCTION, MEASURED IN \$
- TOBACCO - SALES, MEASURED IN \$
- VEGETABLE TOTALS, INCL SEEDS & TRANSPLANTS, IN THE OPEN - SALES, MEASURED IN \$
- WHEAT - SALES, MEASURED IN \$

Select “TOTAL” from the Domain box; select “STATE” from the Geographic Level box; select 2017 from the Year box.

Silage and Haylage Production Value

- Files:
 - [NASS_Silage_Data_2017.csv](#)
 - [NASS_Haylage_Data_2017.csv](#)
 - [NASS_Silage_Data_1519.csv](#)
 - [NASS_Haylage_Data_1519.csv](#)

We used survey data on silage and haylage production quantities from the 2017 Census and prices received from the *Agricultural Prices* report to compute silage and haylage production values for 2017 (U.S. Department of Agriculture, National Agricultural Statistics Service (2019); National Agricultural Statistics Service (2018b)). We also used a range of silage and haylage data from 2015-2019 for the regression analysis described in Appendix C. These data came from the *Crop Production Annual Summary* report and *Agricultural Prices* report. (U.S. Department of Agriculture, National Agricultural Statistics Service (2015a); U.S. Department of Agriculture, National Agricultural Statistics Service (2015b)).

To download the data in the file [NASS_Silage_Data_2017.csv](#) from Quick Stats, select “FIELD CROPS” from the Group box; select “CORN” and “SORGHUM” from the Commodity box; select “PRICE RECEIVED” and “PRODUCTION” from the Category box; select “CORN, GRAIN - PRICE RECEIVED, MEASURED IN \$ / BU”, “CORN, SILAGE - PRODUCTION, MEASURED IN TONS”, and “SORGHUM, SILAGE - PRODUCTION, MEASURED IN TONS” from the Data Item box; select “STATE” from the Geographic Level box; select 2017 from the Year box; select “ANNUAL” from the Period type box. *Note:* this search will also return survey data for corn and sorghum silage production quantities, but we did not use these data in the analysis.

Downloading the data in the file [NASS_Haylage_Data_2017.csv](#) is nearly identical to the above steps. Instead of selecting “CORN” and “SORGHUM” in the Commodity box, select “HAY” and “HAYLAGE”. In the Data Item box, select “HAY - PRICE RECEIVED, MEASURED IN \$ / TON” and “HAYLAGE - PRODUCTION, MEASURED IN TONS”. As with the silage data, this will return survey data for hay production quantities, but we only used the Census data.

For the data in the files [NASS_Silage_Data_1519.csv](#) and [NASS_Haylage_Data_1519.csv](#), first select “SURVEY” in the Program box, then follow all of the same steps outlined above through the Geographic Level box. Then, select 2015-2019 in the Year box and “ANNUAL” in the Period Type box.

Livestock Inventories

- Files:

► [NASS_Livestock_Data_1519.csv](#)

We used a combination of livestock inventory data from the 2017 Census and survey data from the *Cattle* report (U.S. Department of Agriculture, National Agricultural Statistics Service (2019), U.S. Department of Agriculture, National Agricultural Statistics Service ((2015)). For the regression analysis of silage use by dairy cattle in the [4_Silage_Regression.R](#) script, we use the survey data from the *Cattle* report for 2015 through 2019. That way the data are all from a consistent source. The 2017 Census data and survey data on the number of milk replacement heifers in 2017 are used in the [6_Process_Feed_Use_Data.R](#) script.

To download these data from Quick Stats, *do not* make a selection from the Program box; select “ANIMALS & PRODUCTS” from the Sector box; select “LIVESTOCK”, “POULTRY”, and “SPECIALTY” from the Group box; select “CATTLE”, “CHICKENS”, “EQUINE”, “GOATS”, “HOGS”, “SHEEP”, and “TURKEYS” from the Commodity box; select “INVENTORY” from the Category box; select all of the following from the Data Item box:

- CATTLE, COWS, MILK - INVENTORY
- CATTLE, HEIFERS, GE 500 LBS, MILK REPLACEMENT - INVENTORY
- CATTLE, INCL CALVES - INVENTORY
- CATTLE, ON FEED - INVENTORY
- CHICKENS, BROILERS - INVENTORY
- CHICKENS, LAYERS - INVENTORY
- EQUINE, HORSE & PONIES - INVENTORY
- GOATS - INVENTORY
- HOGS - INVENTORY
- SHEEP, INCL LAMBS - INVENTORY
- TURKEYS - INVENTORY

Select “TOTAL” from the Domain box; select “STATE” from the Geographic Level box; select 2015 through 2019 from the Year box; select “POINT IN TIME” from the Period Type box; and select “FIRST OF JAN” and “END OF DEC” from the Period box.

Note: This search will also return survey data for sheep inventories from 2015 to 2019, but these data are not used in the analysis.

Cropland Acreage and Rents

- Files:
 - [CroplandAcresState.csv](#)
 - [CroplandRentState.csv](#)

We used cropland acreage data from the 2017 Census and cropland rent data from the NASS *Cash Rents* survey from 2015 through 2019 (U.S. Department of Agriculture, National Agricultural Statistics Service (2019); U.S. Department of Agriculture, National Agricultural Statistics Service (2015c).

To download the data in the file [CroplandAcresState.csv](#) from Quick Stats, select “CENSUS” from the Program box; select “ECONOMICS” from the Sector box; select “FARMS & LAND & ASSETS” from the Group box; select “AG LAND” from the Commodity box; select “AG LAND, CROPLAND – ACRES” from the Data Item box; select “TOTAL” from the Domain box; select “STATE” from the Geographic Level box; select 2017 from the Year box. Then, we select the columns named Year, State, State ANSI, and Value. The table is ordered by increasing value of the State ANSI variable.

To download the data in the file [CroplandRentState.csv](#) from Quick Stats, select “SURVEY” from the Program box; select “ECONOMICS” from the Sector box; select “EXPENSES” from the Group box; select “RENT” from the Commodity box; select “RENT, CASH, CROPLAND – EXPENSE, MEASURED IN \$/ACRE” from the Data Item box; select “STATE” from the Geographic Level box; select 2010-2023 from the Year box. Then, we select the columns named Year, State, State ANSI,

and Value. The table is ordered by increasing value of the State ANSI variable. *Note:* this search returns data for 2010-2023, but we only used data from 2015-2019 in the analysis.

2.7 COMMODITY FLOW SURVEY AND FREIGHT ANALYSIS FRAMEWORK DATA

- Files:
 - [Special Tab 10.txt](#)
 - [Special Tab 10 UC Davis_Read_Me.pdf](#)
 - [FAF_Shipments_All_2017.csv](#)

We used data from the Commodity Flow Survey and Freight Analysis Framework to calculate dairy product shipments between regions (United States Department of Transportation, Bureau of Transportation Statistics: Office of Survey Programs (2020); United States Department of Transportation: Office of the Secretary of Transportation and Bureau of Transportation Statistics: Office of Statistical and Economic Analysis (2017)). While the public use microdata file is available through the [Bureau of Transportation Statistics repository](#), we requested a special tabulation from the Bureau of Transportation Statistics that includes commodity group observations at the [5-digit Standard Classification of Transported Goods \(SCTG\)](#) commodity code level (United States Department of Transportation, Bureau of Transportation Statistics (2022)). We include the README file prepared for the special tabulation in the [Other_Sources](#) subfolder. The Freight Analysis Framework data are available to download from the [Bureau of Transportation Statistics repository](#).

2.8 UNITED STATES CENSUS BUREAU DATA

We used data from the U.S. Census Bureau on state population and imports and exports of dairy products and feed crops.

Vintage 2019 Population Estimates

- Files:
 - [Census_Population_2017.csv](#)

We used state-level population data for 2017 from the Vintage 2019 Population Estimates (United States Department of Commerce, U.S. Census Bureau (2019)). These data are most easily accessed directly through the [Census Bureau's FTP server](#). We used Table 1. Annual Estimates of the Resident Population for the United States, Regions, States, and Puerto Rico: April 1, 2010 to July 1, 2019 (NST-EST2019-01). The data are also available through the [Census API](#). We edited the CSV file included in this replication package to include only the state name, corresponding FMMO region, and 2017 population estimate.

Import and Export Data

- Files:
 - [Census_Dairy_Product_Exports_2017.csv](#)
 - [Census_Dairy_Product_Imports_2017.csv](#)
 - [GATS_Crop_Exports_1519.csv](#)

We used state-level data on dairy product imports and exports and U.S. total feed crop exports published by the U.S. Census Bureau (United States Department of Commerce, U.S. Census Bureau (2018)). While these data are a Census Bureau product, we accessed the data through the USDA Foreign Agricultural Service [Global Agricultural Trade System \(GATS\)](#).

To download the state-level dairy product export data, start from the GATS homepage and select “Standard Query”; under Data Source, change the dropdown to “U.S. States”; under the Product Groups dropdown select “Ag Sectors (HS-6)-*”; under Reporters select “United States” and “All States”; under Partners select “World Total”; under Products select “Dairy Products-*” and click “List HS6 Codes”; select all products listed in the box that appears; under Statistics, change the measure of Value from Thousands to Dollars; under Dates select Annual, Jan-Dec, 2017, and Dec for End Month; under Settings change Precedence to “US State/Product/Partner”, change Order By to “Code”, and change Calculation to “(None)”;

click Retrieve Data, then click Create CSV File. To download the state-level dairy product import data, keep the same selections but change the Product Type dropdown to “Imports”.

To download the U.S. total feed crop export data, start from the GATS homepage and select “Standard Query”; under the Product Groups dropdown select “Harmonized (HS-6)”; under Partners select “World Total”; under Products select “10 - Cereals-” and “12 - Oil Seeds and Oleaginous Fruits-” and click List Selected Chapters; select all of the following harmonized codes:

- 100119 - Durum Wheat
- 100199 - Wheat, except Durum
- 100290 - Rye
- 100390 - Barley
- 100490 - Oats
- 100590 - Corn (Maize), except Seed
- 100610 - Rice in the Husk
- 100620 - Rice, Husked (Brown)
- 100630 - Rice, Semi-Milled
- 100640 - Rice, Broken
- 100790 - Grain Sorghum
- 100810 - Buckwheat
- 100829 - Millet
- 100840 - Fonio
- 100850 - Quinoa
- 100860 - Triticale
- 100890 - Cereals Nesoi, Including Wild Rice
- 120190 - Soybeans
- 120400 - Flaxseed (linseed)
- 120510 - Rapeseed, Low Erucic Acid
- 120590 - Rapeseed, Nesoi
- 120729 - Cottonseed
- 120799 - Oilseeds Nesoi
- 121410 - Alfalfa Meal & Pellets
- 121490 - Forage Products (Hay, Clover, etc.)

Under Statistics, change the measure of Value from Thousands to Dollars; under Dates select Annual, Jan-Dec, 2015-2019, and Dec for End Month; under Settings change Order By to “Code” and change Calculation to “(None)”; click Retrieve Data, then click Create CSV File.

2.9 USDA ECONOMIC RESEARCH SERVICE DATA

Feed Grains Database, Oil Crops Yearbook, and U.S. Bioenergy Statistics

- Files:
 - [feed-grains-yearbook-all-years.csv](#)
 - [OilCropsAllTables.csv](#)
 - [us-bioenergy-statistics.csv](#)

We used data from the ERS Feed Grains Database Yearbook Tables, the Oil Crops Yearbook, and the U.S. Bioenergy Statistics (U.S. Department of Agriculture, Economic Research Service (2026a); U.S. Department of Agriculture, Economic Research Service (2026b); U.S. Department of Agriculture, Economic Research Service (2026c). To access the Feed Grains

Database, go to the [Feed Grains Database](#) website and download the CSV version of the file “Feed Grains Yearbook Tables - All years”. We used tables 4-7 and 30.

To access the Oil Crops Yearbook data, go to the [Oil Crops Yearbook](#) website and download the CSV version of the file “All Tables - Oil Crops Yearbook”. We used tables 3-5, 18-20, 22-27, and 29-31.

To access the U.S. Bioenergy Statistics data, go to the [U.S. Bioenergy Statistics](#) website and download the CSV version of the file “U.S. Bioenergy Statistics”. We used tables 8.1, 8.2, and 8.3.

Dairy Data

- Files:
 - [selected-dairy-products-percapita.csv](#)
 - [ERS_Component_Conversions.csv](#)

We used data from the ERS Dairy Data database on per capita consumption of beverage milk products and component conversion factors for dairy products (U.S. Department of Agriculture, Economic Research Service (2026d). To access the per capita consumption data, go to the [Dairy Data](#) website, click on “Current dairy data”, and download the CSV version of the file “Dairy products: Per capita consumption, United States (Annual)”. To access the component conversions data, go to the [Dairy Data Documentation](#) website, scroll to the section titled “Dairy Data Documentation Supporting Files”, and download the file “Supply-allocation-conversion.csv”. The CSV file included in this replication package was edited by the authors to eliminate unnecessary information before use.

2.10 LOOKUP TABLES

We created several lookup tables to facilitate matching values between different datasets. For example, matching state names to FMMO regions. These lookup tables are located in the “Lookup_Tables subfolder within the”Data” folder. The contents of the folder are listed in Table 2.

3 DATASET LIST

Table 1: List of Files in Data/Raw Folder

File Path	Size	Source	Pro- vided
Data/Raw/California_Monthly_ClassI_2017.csv	179	CDFA	TRUE
Data/Raw/California_Statistics_2017.csv	371	CDFA	TRUE
Data/Raw/Census_Dairy_Product_Exports_2017.csv	74.23K	U.S. Census Bureau	TRUE
Data/Raw/Census_Dairy_Product_Imports_2017.csv	43.16K	U.S. Census Bureau	TRUE
Data/Raw/Census_Population_2017.csv	1.66K	U.S. Census Bureau	TRUE
Data/Raw/CroplandAcresState.csv	1.5K	USDA-NASS	TRUE
Data/Raw/CroplandRentState.csv	13.9K	USDA-NASS	TRUE
Data/Raw/Datamart-Export_DY_Advanced_Prices_2017.csv	573	USDA-AMS	TRUE
Data/Raw/Datamart-Export_DY_Class_Prices_2017.csv	1.18K	USDA-AMS	TRUE

File Path	Size	Source	Pro- vided
Data/Raw/Datamart-Export_FM_ClassIII_Util-Butterfat_2017.csv	1.1K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassIII_Util-Other Solids_2017.csv	783	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassIII_Util-Protein_2017.csv	783	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassIII_Util-Utilization_2017.csv	1.21K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassII_Util-Butterfat_2017.csv	1.16K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassII_Util-NFS_2017.csv	783	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassII_Util-Utilization_2017.csv	1.2K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassIV_Util-Butterfat_2017.csv	1.11K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassIV_Util-NFS_2017.csv	783	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassIV_Util-Utilization_2017.csv	1.2K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassI_Util-Butterfat_2017.csv	1.1K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassI_Util-Milk_2017.csv	1.05K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassI_Util-NFS_2017.csv	783	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassI_Util-Utilization_2017.csv	1.24K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_Producer_Components-Butterfat_2017.csv	1.1K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_Producer_Components-Other Solids_2017.csv	783	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_Producer_Components-Protein_2017.csv	783	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_Producer_Receipts_2017.csv	1.29K	USDA-AMS	TRUE
Data/Raw/ERS_Component_Conversions.csv	323	USDA-ERS	TRUE
Data/Raw/FAF_Shipments_All_2017.csv	578.45K	U.S. Census Bureau	TRUE
Data/Raw/feed-grains-yearbook-all-years.csv	16.22M	USDA-ERS	TRUE
Data/Raw/GATS_Crop_Exports_1519.csv	2.54K	USDA-FAS	TRUE
Data/Raw/MilkAndMilkPooledByState2017.csv	4.25K	USDA-AMS	TRUE
Data/Raw/MilkAndMilkPooledByState2017.xlsx	25.11K	USDA-AMS	TRUE
Data/Raw/MilkAndMilkPooledByState2019.csv	4.46K	USDA-AMS	TRUE
Data/Raw/MilkAndMilkPooledByState2019.xlsx	25.61K	USDA-AMS	TRUE
Data/Raw/NASS_All_Cropland_2017.csv	12.36K	USDA-NASS	TRUE
Data/Raw/NASS_All_Cropland_Harvested_2017.csv	6.48K	USDA-NASS	TRUE
Data/Raw/NASS_Census_Feed_Crop_Acreage_2017.csv	62.2K	USDA-NASS	TRUE
Data/Raw/NASS_Census_Other_Crop_Acreage_2017.csv	49.77K	USDA-NASS	TRUE
Data/Raw/NASS_Crop_Sales_2017.csv	58.6K	USDA-NASS	TRUE
Data/Raw/NASS_Dairy_Products_2017.csv	19.78K	USDA-NASS	TRUE

File Path	Size	Source	Pro- vided
Data/Raw/NASS_Haylage_Data_1519.csv	41.79K	USDA-NASS	TRUE
Data/Raw/NASS_Haylage_Data_2017.csv	12.45K	USDA-NASS	TRUE
Data/Raw/NASS_Livestock_Data_1519.csv	195.55K	USDA-NASS	TRUE
Data/Raw/NASS_Milk_Production_2017.csv	6.22K	USDA-NASS	TRUE
Data/Raw/NASS_Silage_Data_1519.csv	69.22K	USDA-NASS	TRUE
Data/Raw/NASS_Silage_Data_2017.csv	17.72K	USDA-NASS	TRUE
Data/Raw/NASS_Total_Crop_Sales_2017.csv	6.76K	USDA-NASS	TRUE
Data/Raw/OilCropsAllTables.csv	7.65M	USDA-ERS	TRUE
Data/Raw/selected-dairy-products-percapita.csv	69.9K	USDA-ERS	TRUE
Data/Raw/Special Tab 10.txt	148.8M	U.S. Census Bureau	TRUE
Data/Raw/State_Region_Match_Table.csv	1.17K	NA	TRUE
Data/Raw/us-bioenergy-statistics.csv	3.32M	USDA-ERS	TRUE
Data/Raw/Western_New_York_Data_2017.csv	175	NYDAM	TRUE
Data/Raw/Western_New_York_Utilization_2017.csv	92	NYDAM	TRUE

Table 2: List of Files in Data/Lookup_Tables Folder

File Path	Size	Source	Provided
Data/Lookup_Tables/Census_Regions_Divisions_States.csv	2.43K	Created by the authors	TRUE
Data/Lookup_Tables/Class1_Differentials.csv	206	Created by the authors	TRUE
Data/Lookup_Tables/Crop_Groups.csv	602	Created by the authors	TRUE
Data/Lookup_Tables/HSCode_Class.csv	530	Created by the authors	TRUE
Data/Lookup_Tables/HSCode_Crops.csv	528	Created by the authors	TRUE
Data/Lookup_Tables/NASS_Dairy_Product_Match_Table.csv	607	Created by the authors	TRUE
Data/Lookup_Tables/NASS_Livestock_Match_Table.csv	494	Created by the authors	TRUE
Data/Lookup_Tables/NASS_Regions_Match_Table.csv	891	Created by the authors	TRUE
Data/Lookup_Tables/State_Region_Match_Table.csv	1.26K	Created by the authors	TRUE

Table 3: List of Files in Data/Other_Sources Folder

File Path	Size	Source	Provided
Data/Other_Sources/California_Dairy_Statistics_Annual_2017.pdf	5.58M	C DFA	TRUE
Data/Other_Sources/Special Tab 10 UC Davis_Read_Me.pdf	129.63K	BTS	TRUE
Data/Other_Sources/Western_New_York_Statistics_Annual_2017.pdf	538.42K	NY-DAM	TRUE

4 COMPUTATIONAL REQUIREMENTS

We used R version 4.5.2 and the packages listed in Table 4 running in RStudio version 2026.7.1+147 to process the raw data contained in [Data/Raw](#) for analysis (R Core Team (2025); Posit team (2026)). We used GAMS version 54.2.1 and the [CONOPT3](#) solver for the main model. We used R to process the results from the main model to produce the tables in the manuscript and two figures in the supplemental material. Additionally, we used Wolfram Mathematica version 12.2.0.0 to produce the figures in the manuscript and additional figures in the supplemental material.

The [run_all.sh](#) script can be executed in the Terminal on a computer running Mac OS, but some additional setup is required on a computer running Windows. We used Git Bash to execute the [run_all.sh](#) script on a computer running Windows. Git Bash is installed with Git for Windows, so it is included in the list of necessary software below.

List of Necessary Software:

- R 4.5.2
 - [R for Windows](#)
 - [R for MacOS](#)
- RStudio 2026.07.1+147 (Optional)
- GAMS Version 54.2.1
 - [CONOPT3](#) solver
- GAMS Studio Version 1.25.2 (Optional)
- Wolfram Mathematica Version Number 12.2.0.0
- Necessary Software for Windows Systems:
 - [Git for Windows](#)

Table 4: R Packages Used

Package	Version	Citation
base	4.5.2	R Core Team ((2025))
fs	2.1.0	Hester, Wickham, and Csárdi ((2026))
grateful	0.3.0	Rodriguez-Sanchez and Jackson ((2025))
here	1.0.2	Müller ((2025))
knitr	1.51	Xie ((2014)); Xie ((2015)); Xie ((2025))
plyr	1.8.9	Wickham ((2011))
renv	1.2.4	Ushey and Wickham ((2026))
rmarkdown	2.31	Xie, Allaire, and Golemund ((2018)); Xie, Dervieux, and Riederer ((2020)); Allaire et al. ((2026))
sf	1.1.2	Pebesma ((2018)); Pebesma and Bivand ((2023))
stargazer	5.2.3	Hlavac ((2022))
tidyverse	2.0.0	Wickham et al. ((2019))

4.1 MEMORY, RUNTIME, AND STORAGE REQUIREMENTS

Summary

Approximate time needed to reproduce the analyses on a standard 2026 desktop machine:

- Windows: 3 minutes
- MacOS: 10 minutes

Approximate storage space needed:

- 233 MB (when initially unzipped)
- 302 MB (with necessary R packages installed)

Details

The data processing portions of the code were last run on a 6-core AMD-based desktop with 32 GB of RAM and 475 GB storage (45.8 GB free) running Windows 11 version 25H2.

The main model code was last run on a 14-inch 2021 Apple Macbook Pro with an Apple M1 chip and 32 GB of RAM running Mac OS Sonoma 14.7.

When the `run_all.sh` script is executed, the `0_Data_Processing_Run_All.R` script will load R, execute the code in the `renv/activate.R` script, and run a `renv::restore()` command to install all necessary R packages. The code in the `renv/activate.R` script will bootstrap `{renv}`, then `renv::restore()` will install the packages listed in the `renv.lock` file. On a Windows computer, this process takes approximately 2 minutes since all packages are able to be installed from binary files. On MacOS, the process takes approximately 10 minutes since some packages need to be installed and compiled from source. After the first time `run_all.sh` is run, the `renv::restore()` command will not need to install packages so the execution time will be faster.

5 DESCRIPTION OF CODE SCRIPT FILES

5.1 R CODE

Data Processing

- `0_Data_Processing_Run_All.R`
 - Runs `renv::restore()` to install necessary R packages; creates destination folders for intermediate and analysis data files; uses the `source()` function to run all other R scripts in the `Data-Processing` subfolder.
- `1_Data_Import.R`
 - Imports all data in the `Data/Raw` folder and creates two vectors of continental state names and FMMO region names.
- `2_Process_Milk_Shipments.R`
 - Processes the milk utilization by state of origin data.
 - Output:
 - `GAMS_CODE/CSV_DATA_FILES/Milk_Shipments.csv`
 - `Data/Processed/Milk_Production.csv`
 - `Data/Processed/Milk_Utilization.csv`
 - `Data/Processed/Milk_Not_Pooled.csv`
- `3_Process_Component_Values_Volumes.R`
 - Processes data on milk utilization by class to calculate the volumes and values of milk component by class.
 - Output:
 - `GAMS_CODE/CSV_DATA_FILES/Comp_Quantity_for_GAMS.csv`
 - `GAMS_CODE/CSV_DATA_FILES/Comp_Quantity_for_GAMS.csv`
- `4_Process_CFS_Data.R`
 - Processes the Commodity Flow Survey data.
 - Output:
 - `GAMS_CODE/CSV_DATA_FILES/CFS_Dairy_for_GAMS_Gravity.csv`
- `5_Silage_Regressions.R`
 - Estimates regional regressions for silage use by dairy cattle and cattle on feed to calculate the share of silage used by dairy cattle.

- Output:
 - `Data/Processed/Dairy_Silage_Share.csv`
- `6_Process_Feed_Use_Data.R`
 - Uses data from the ERS Feed Grains Database, Oil Crops Yearbook, and U.S. Bioenergy Statistics to calculate the share of each feed crop used by the dairy sector.
 - Output:
 - `Data/Processed/State_FCAU_Shares.csv`
 - `Data/Processed/Crop_Feed_Shares.csv`
- `7_Process_NASS_Crop_Data.csv`
 - Processes the cropland acreage and production value for each feed crop and calculates the share utilized by the dairy sector.
 - Output:
 - `GAMS_CODE/CSV_DATA_FILES/Crop_Areas.csv`
 - `GAMS_CODE/CSV_DATA_FILES/Crop_Prod_Values.csv`
 - `GAMS_CODE/CSV_DATA_FILES/Dairy_Crop_Cons_Values.csv`
- `8_CroplandRentQE.R`
 - Calculates the cropland rent for each FMMO region.
 - Output:
 - `GAMS_CODE/CSV_DATA_FILES/Cropland_Rent.csv`

Tables

- `TablesQE.R`
 - Uses the output from the GAMS model to produce the tables for the manuscript and supplemental material.
 - See Table 5 for the tables produced by this file.

Figures

- `MapsFMMOQE.R`
 - Produces two maps of FMMO regions, one depicting the actual regional boundaries and one depicting the approximations used in this analysis, included in the supplemental material.

Claims

- `TextClaimsQE.R`
 - Uses the output from the GAMS model to support the in-text claims in the manuscript and supplemental material.

5.2 GAMS CODE

- `FMMO_Model_QE.gms`
 - Contains the main analysis model used to determine the results under the main and alternative counterfactual scenarios.

5.3 MATHEMATICA CODE

- `FMMOQE.wls`
 - Produces figures in the manuscript and the supplemental material. Table 5 lists the specific figures produced by this code.

5.4 LICENSE FOR CODE

The code included in this replication package is licensed under the Apache 2.0 license. A copy of the license may be obtained at <https://www.apache.org/licenses/LICENSE-2.0>. See `LICENSE.txt` for more information.

6 INSTRUCTIONS TO REPLICATORS

The code for data processing, analysis, and table/figure creation is all executed through a shell script called `run_all.sh`. The following steps explain how to execute the `run_all.sh` script.

1. Install all necessary software.
2. Unzip the `Geographic-Footprint-US-Dairy.zip` folder in a designated location.
3. Open a Terminal or Git Bash instance:
 - a. On Windows, use Windows Explorer to navigate to the `Geographic-Footprint-US-Dairy` folder. Right click within the folder and select “Open Git Bash here” (this may be listed under the “Show more options” menu in Windows 11).
 - b. On MacOS, open Terminal then use the `cd` command to change the working directory to the `Geographic-Footprint-US-Dairy` folder.
4. Execute the command `bash run_all.sh`.

We believe the execution instructions for MacOS would also apply for a Linux system, but we did not have an opportunity to test the code on a computer running Linux.

Once the `run_all.sh` script has successfully executed, the main folder will contain three new subfolders, MANUSCRIPT CLAIMS, MANUSCRIPT FIGURES, and MANUSCRIPT TABLES, that will contain the replicated tables, figures, and in-text claims from the manuscript. The specific code file that creates each table or figure is listed in Table 5.

7 LIST OF TABLES AND FIGURES AND ASSOCIATED CODE FILES

Table 5: Code Files to Generate Tables and Figures in Manuscript

Figure/Table	Script File	Output File
Table 1	TablesQE.R	table1.txt
Table 2	TablesQE.R	table2.txt
Table 3	TablesQE.R	table3.txt
Table 4	TablesQE.R	table4.txt
Table 5	TablesQE.R	table5.txt
Table 6	TablesQE.R	table6.txt
Table 7	TablesQE.R	table7.txt
Table C.2	3_Process_Component_Values_Volumes.R	Table_C2.txt
Table C.3	6_Process_Feed_Use_Data.R	Table_C3.txt
Table C.4	6_Process_Feed_Use_Data.R	Table_C4.txt
Table C.5	5_Silage_Regression.R	Table_C5.txt
Table C.6	5_Silage_Regression.R	Table_C5.txt
Table D.1	TablesQE.R	tabled1.txt
Table D.2	TablesQE.R	tabled2.txt
Table D.3	TablesQE.R	tabled3.txt
Table D.4	TablesQE.R	tabled4.txt
Table E.1	TablesQE.R	tablee1.txt

Figure/Table	Script File	Output File
Figure 1	FMMOQE.wls	figure1.png
Figure A.1	MapsFMMOQE.R	fmmoregion.png, fmmoapproximations.png
Figure G.1	FMMOQE.wls	figureg1.png
Figure G.2	FMMOQE.wls	figureg2.png
Figure G.3	FMMOQE.wls	figureg3.png
Figure G.4	FMMOQE.wls	figureg4.png
Figure G.5	FMMOQE.wls	figureg5.png

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